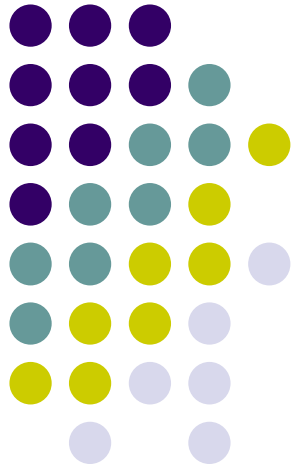

بسم الله الرحمن الرحيم

Disorders of Sebaceous Glands

Aya Yousef Badran

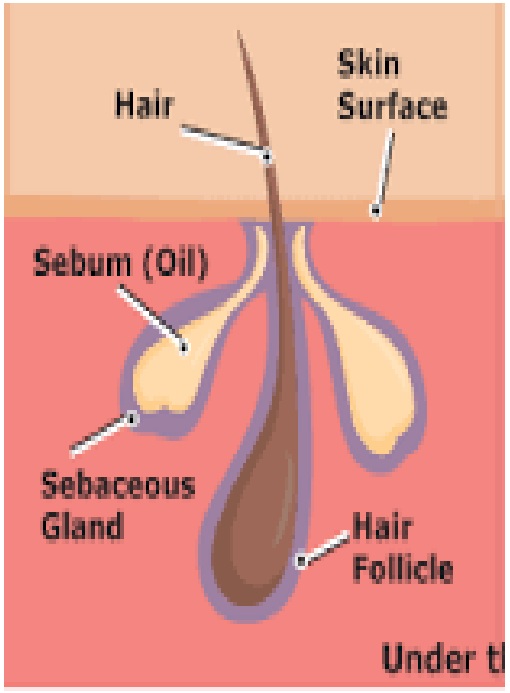
Assistant Professor of Dermatology

Assiut University Hospital



Sebaceous gland

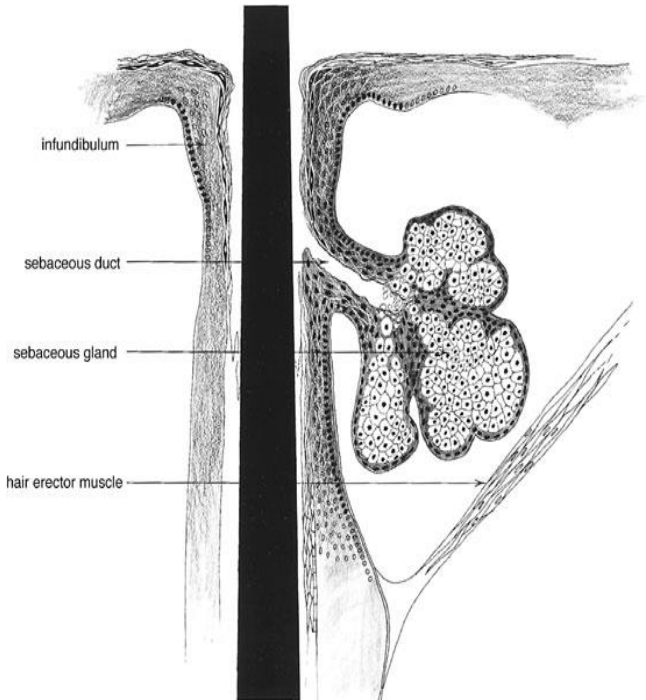
- Present in **on all skin sites** **except** the palms and soles
- found at great abundance on **face** and scalp.



- Each gland **open into a hair follicle** (pilosebaceous unit).

- Start their activity after **puberty.**

- They are under **androgenic control.**



- Functions of sebum:
  - Moisturizing the skin.
  - Protects the skin against fungal and bacterial infection.

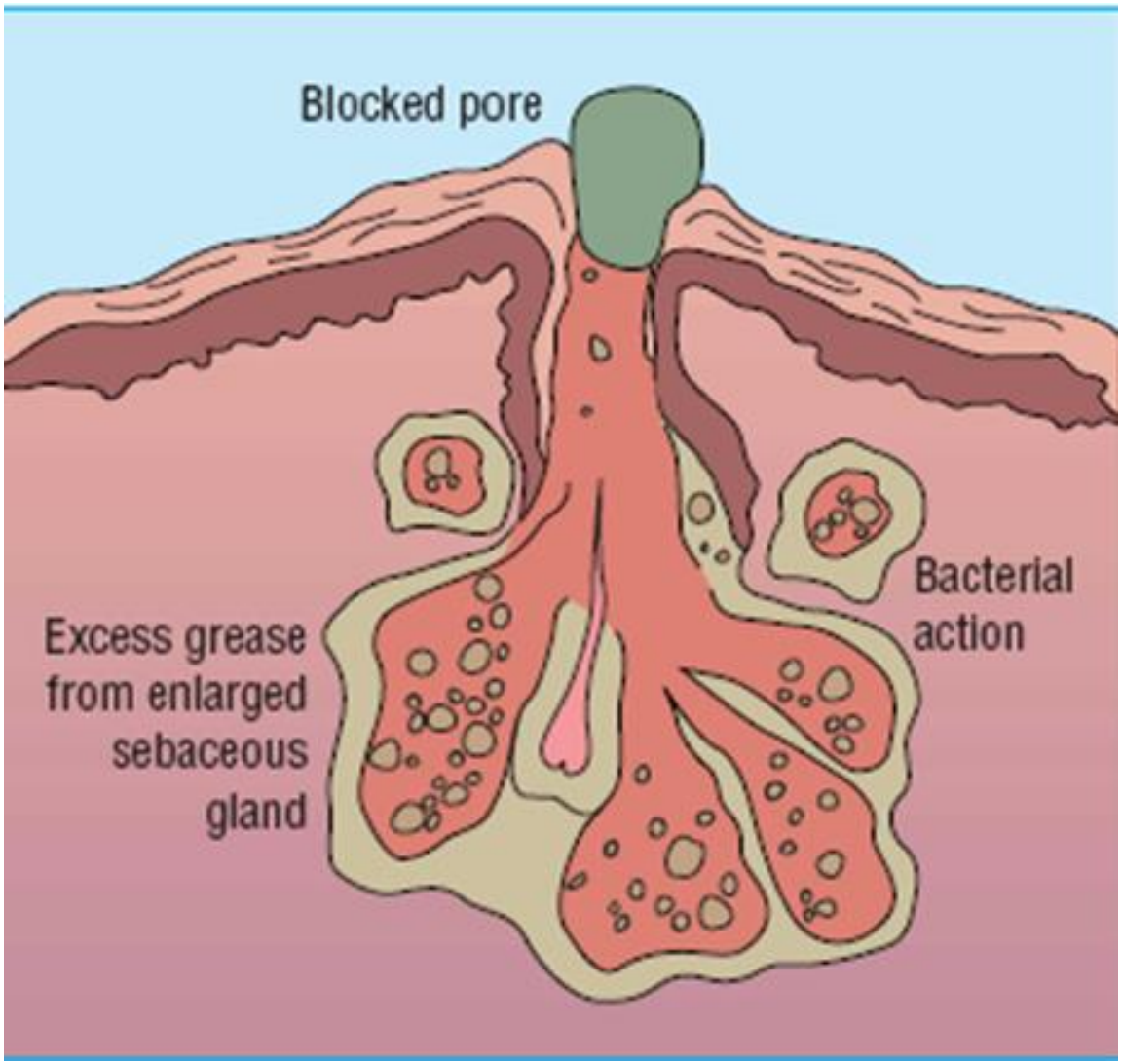
Acne Vulgaris

Def.: chronic inflammatory disease of the pilosebaceous units.

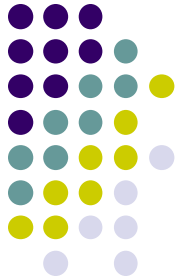
Causes:

- **Androgenic** stimulation of sebaceous glands → increased sebum production
- **Ductal** hyper-cornification
- Proliferation of microbial flora (**Cutibacterium acnes**).
- **Inflammation**.

Distension of the follicles with tightly packed horny cells comedo.



Factors which aggravate acne



- Menstruation
- Stress.
- Sweating
- Cosmetics & hair pomade.
- Medications: iodides, antiepileptics, steroids.

Clinical picture:

- Occurs in adolescents & young adults
- Occurs mostly on **face** (cheeks mostly and less on forehead , nose and chin) and less frequently on **back and chest**.

Clinical Picture

Acne Lesions

polymorphic (one type may predominate) .





Acne lesions

- **Comedones**
- **Papules**
- **Pustules.**
- **Nodules and cysts.**

- **Post-inflammatory lesions**

NON-INFLAMED LESIONS

Comedones

- **The the characteristic lesion of acne.**

- **Blackheads** (open comedones): when follicular orifice is opened + distended.

- **Whiteheads** (closed comedones), accumulated sebum and keratin plug pilosebaceous duct.

Black Heads (open comedones)





White Heads (closed comedones)



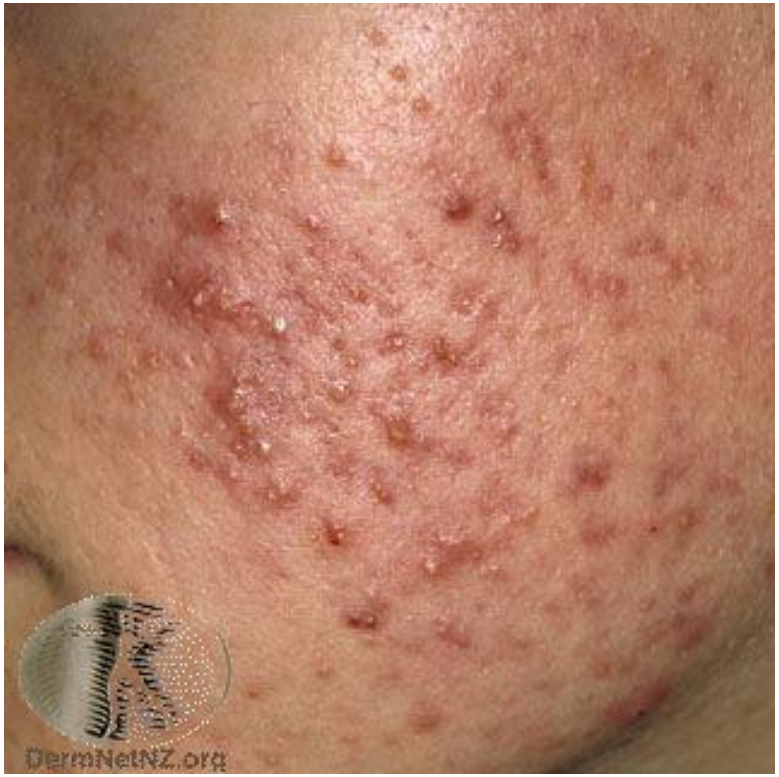


INFLAMED LESIONS

- **Papules**, small red papules.
- **Pustules**, contain pus.
- **Nodules and cysts**, large pus-filled painful acne lesion.

Papule and Pustules





Nodules and Cysts



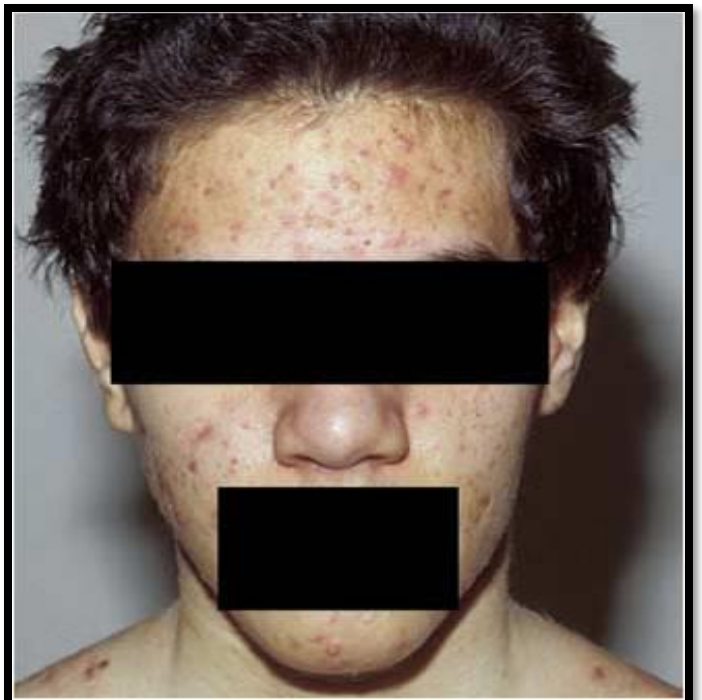












Post –inflammatory lesions

- Erythema.
- Pigmentation.
- Scarring (atrophic or keloidal).

Post-acne Scars

- Small atrophic scars or pits, deeper furrows.

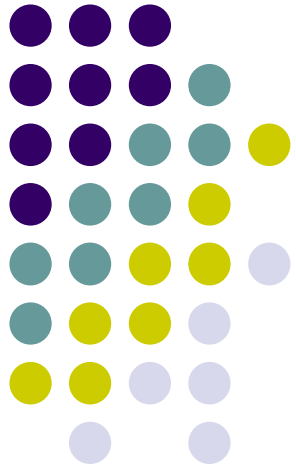


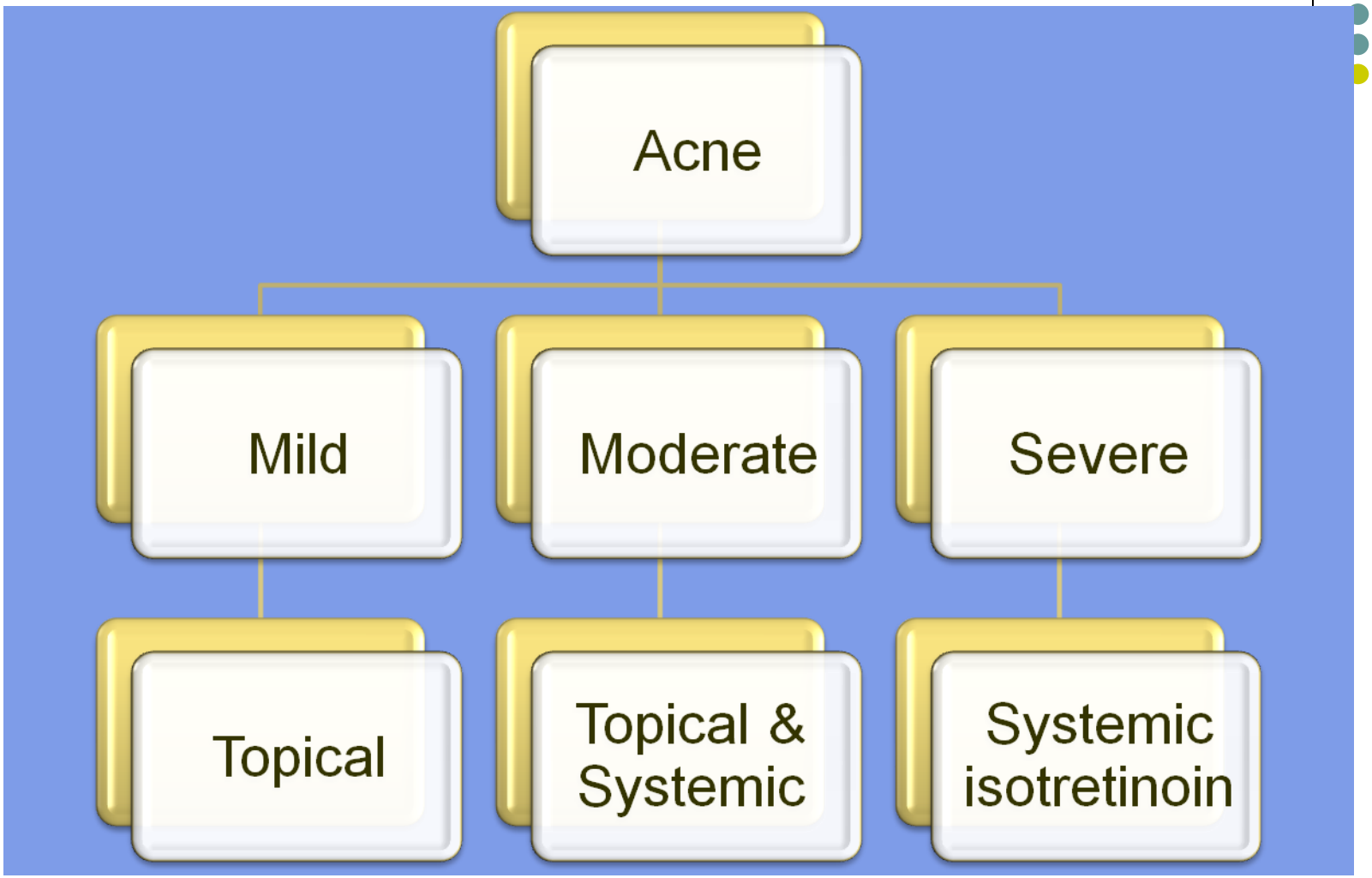


Hypertrophic/keloid scars



TREATMENT OF ACNE




Acne
Mild
Moderate
Severe
Topical
Topical & Systemic
Systemic isotretinoin

Lines of treatment:

- Topical treatment.
- Systemic Treatment.
- Physical modalities.
- Treatment of post-acne scars

Topical treatment

- **Topical antibiotics** e.g. erythromycin, clindamycin
- **Benzoyl peroxide 5%.**
- **Retinoids** e.g. retinoic acid (0.025-0.05%), adapalene
- **Azaleic acid (20%).**
- **Topical Combination**

Systemic Treatment

1. **Antibiotics**
2. **Antiandrogens**
3. **Systemic retinoids**
4. **Systemic steroids**.

1-Oral Antibiotics

- Not more than 3 months

Action: downregulate C. acnes enzymes, inhibition of inflammation

- Tetracyclines- Doxycycline, minocycline

Not used for children <10 years old & in pregnancy.

- Erythromycin: used in pregnant women.
- Others: clindamycin, azithromycin, Trimethoprim/sulfamethoxazole

2- **Antiandrogens** (Cyproterone acetate)

Used only in adult females, contraindicated in males & pregnant women.

3- Systemic retinoids (**isotretinoin**): used in **moderate and severe** acne not responding to conventional treatment. Dose 0.5-1 mg /kg/day,

Side effects: teratogenic, increase serum triglycerides and liver enzymes.

4- Systemic steroids used only in severe cases.



PHYSICAL MODALITIES



- Comedo extraction

- Intralesional injection of steroids.



- Cryotherapy.

TREATMENT OF POSTACNE SCARS AND HYPERPIGMENTATION





-**Chemical peeling**

-**Microdermabrasion**

-**Laser skin resurfacing** (**Fractional laser**)

-**Micro-needling**

-**Platelet rich plasma**(**PRP**)




Thank you
Thank you